Solution Document for Gate Operating System (GOS)

Adani Port Ennore

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1 Axestrack Overview

Axestrack stands as a pioneer in transforming the logistics industry with innovative solutions that redefine fleet management. With over **15 years of expertise** and only Indian company ever to be recognized in **Gartner's Market Guide for Fleet Telematics** Solutions, we are a trusted name for businesses seeking smarter, safer, and more efficient operations.

At the core of our offerings is the **Axestrack Operating System (aOS)**, a next-generation AI platform designed to go beyond traditional tracking. Built on advanced analytics, intelligent automation, and real-time insights, aOS delivers comprehensive solutions to optimize fleet productivity, reduce TAT, providing end-to-end visibility and control for inbound, in-plant, outbound and port operations and improve operational efficiency. (For more about the platform please visit - https://www.axestrack.com/)

Axestrack is more than a technology provider—we are a transformation partner for plant logistics. Through **CPL (Connected Plant Logistics),** we help enterprises digitize and **orchestrate inbound movements, gate operations, yard management, in-plant logistics, and outbound dispatch** as one connected flow. CPL brings together real-time visibility, actionable insights, and intelligent control across people, vehicles, assets, and processes—supported by smart integrations with ERP/WMS/TMS and partner systems. This enables operations teams to predict bottlenecks, prevent exceptions, and resolve issues early—reducing TAT, improving compliance, and ensuring smoother, faster, and more reliable plant-to-network execution..

Recognized globally and trusted locally, Axestrack is committed to shaping the future of logistics. We are investing in advanced telematics, IoT integrations, and cloud-based control towers to ensure our clients stay ahead in an ever-evolving landscape.

With Axestrack, every journey becomes smarter, safer, and future-ready.

Few of our successful customers

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Proven Value Delivered to Market Leaders

Axestrack has consistently delivered measurable value to market-leading organizations, including:

- Better dispatch reliability and SLA adherence
- Lower cost via reduced idle time and detention
- Higher productivity through faster TAT
- Stronger safety, compliance, and auditability
- End-to-end visibility across all stakeholders

By integrating fleet telematics, logistics execution, and advanced analytics, Axestrack enables enterprises to transform logistics from a reactive cost function into a **strategic, performance-driven capability**.

CPL (Connected Plant Logistics) – End-to-End Plant & Yard Logistics Orchestration Platform

CPL is Axestrack's enterprise platform purpose-built to digitize and orchestrate end-to-end plant logistics—from inbound vehicle scheduling and gate entry to yard movement, in-plant operations, and outbound dispatch. Designed for high-throughput industrial environments, CPL connects plants, transporters, warehouses, and enterprise systems into a single operational layer, delivering real-time control, faster turnaround, and measurable productivity gains.

Key capabilities include:

- Inbound scheduling / slotting to reduce congestion
- Gate automation and digital documentation
- Smart yard & in-plant movement visibility
- TAT tracking with bottleneck insights
- Exception alerts for delays and risks
- Role-based dashboards for operations and leadership

CPL supports modern logistics operating models by enabling enterprises to shift from manual coordination to data-led, exception-driven plant logistics management—improving throughput, reducing idle time, strengthening governance, and delivering predictable, efficient plant-to-network execution.

Enterprise-Grade Security, Scalability, and Governance

All Axestrack platforms are built with enterprise-grade security, scalability, and reliability at their core. This includes role-based access control, auditability, high availability, and compliance-ready architectures—ensuring suitability for large, mission-critical deployments.

Partnership-Led Engagement Model

Axestrack follows a collaborative delivery model, working closely with client stakeholders to ensure successful rollout, adoption, and continuous value realization. This includes structured onboarding, configuration aligned to operating models, and continuous optimization driven by operational data and business outcomes.

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2 Business Requirements Summary

2.1 As-Is Process – Export and Import Operations

Currently, logistics operations at Adani Ennore Port are largely manual and supported by multiple standalone systems without a unified workflow. In the export process, before a vehicle arrives at the gate, a Port Entry Permit (PEP) is expected to be created in the ITUP system. If the PEP is not pre-created, the vehicle enters the parking area and generates the PEP through a kiosk available onsite. Simultaneously, the N4 Navis system provides vehicle and container details. In parallel, another system called Universal Sealing System captures seal details and vehicle images from multiple angles. All these details are printed and submitted physically to the customs officer for review and approval. In certain cases, customs may grant partial approval, allowing the vehicle to exit the parking area but restricting it from loading material to the vessel until re-verification and final approval is completed.

Once final approval is granted, the vehicle enters the port and terminal gate, completes unloading, and exits the port, concluding the export process. The import process follows a similar flow where trucks enter the parking area with an already generated PEP and proceed to the terminal for loading. Post service, customs officers manually verify physical documents and provide approval before the vehicle exits the port. Gate movements across the process are handled manually, and there is no automated timestamp capture for vehicle entry, exit, or dwell time tracking.

Overall, the process is fragmented across two to three standalone systems with no centralized or integrated platform for customs officers to digitally verify, approve, or track vehicle and container movements.

2.2 Major Business Pain Points

- **Manual Dependency & Paper-Based Approvals**
  Heavy reliance on physical documents and manual verification increases processing time and risk of human error.
- **Lack of System Integration**
  Multiple standalone systems (ITUP, N4, Universal Sealing System) operate in silos with no unified dashboard for end-to-end visibility.
- **No Real-Time Tracking or Timestamping**
  Absence of automated gate timestamps prevents accurate dwell time analysis, SLA monitoring, and performance tracking.
- **Limited Visibility for Customs**
  Customs officers do not have a centralized digital system to verify, cross-check, or provide structured approvals (including partial approvals).
- **Audit & Compliance Risks**
  Manual handling and physical document submission create challenges in audit trails, traceability, and compliance reporting.
- **Scalability Constraints**
  The current process may not efficiently support higher traffic volumes due to manual gate handling and lack of automation.

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3 To-Be Process- Proposed Gate Operating System (GOS)

To address the current operational inefficiencies, a Gate Operating System is proposed that integrates seamlessly with the existing ITUP, N4 Navis, and Universal Sealing system. This platform will consolidate all vehicle, container, seal, and documentation data into a single, centralized system from the moment a vehicle enters the parking area until customs approval is completed. By aggregating and synchronizing data in real time, the system will provide customs officers with a one-stop digital interface to verify all required information, eliminating reliance on physical documents and multiple standalone systems.

The proposed solution will support structured approval workflows, including both full approval and partial approval scenarios. Based on the type of approval granted, the system will automatically trigger the subsequent workflow steps, ensuring controlled movement of vehicles within the port ecosystem. This workflow-driven approach will improve governance, reduce ambiguity, and minimize manual intervention. Additionally, the platform will maintain comprehensive audit trails, digital records, and approval logs, strengthening compliance, traceability, and reporting capabilities while enabling operational scalability as business volumes increase.

In Phase 2, the platform will further integrate with IoT sensors installed at strategic locations such as parking entry/exit, port gate entry/exit, terminal entry/exit and port entry/exit. These integrations will enable automated timestamp capture and real-time vehicle movement tracking. The system will provide stage-wise visibility of each vehicle within the port lifecycle, improving turnaround time monitoring, congestion management, SLA adherence, and data-driven decision-making. Overall, the proposed solution transforms the current manual and fragmented operations into a fully integrated, transparent, and scalable digital ecosystem.

4 Functional Scope

4.1 Export Process

- Develop functionality in the GOS to support both Full and Partial approval options for Customs officers during the export approval process.
- Enable Customs officers to perform export approvals seamlessly using both Mobile and Web applications, providing flexibility in operational environments.
- Update the user interface and approval workflows to present Customs officers with all relevant information required to make informed decisions during the approval process.
- Implement automatic updates to the Hold Release flag in the N4 system, triggered appropriately based on whether the Customs officer grants Full or Partial approval.
- Retrieve and display comprehensive container details for export shipments, ensuring Customs officers have complete visibility when reviewing export approvals.
- Provide Customs officers with a detailed list of Container Freight Station (CFS) movements relevant to the export shipments, helping them verify shipment status during approval.
- Integrate with N4 APIs to fetch CFS movement data filtered by specified date ranges, ensuring timely and accurate information is presented to Customs officers.
- Incorporate e-Seal vendor API integration to retrieve seal details for containers, supporting compliance and security checks during the Customs approval process.

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4.2 Import Process

- Enable Customs officers to review and approve import containers subject to scan holds through both Web and Mobile applications, facilitating faster clearance processes.
- Trigger the release of scan hold flags in the N4 system via API calls once Customs officers grant approval, ensuring container status is accurately updated in real time.
- Implement notifications to the ITUP system at various stages of the import truck's movement through the port, supporting real-time milestone tracking and improved operational coordination.

5 Technical Scope

- Integration with ITUP for PEP
- Integration with N4 to get Container details, hold release and scan hold release
- Integration with E-seal vendor to get the seal and vehicle image details
- The platform is a SaaS based and to be hosted on Axestrack's cloud and separate efforts to be calculated for hosting on Adani cloud.

6 Out of Scope

- **Hardware Installation:** The entire system will be a software layer and hardware installation is out of scope.
- **Civil works for hardware installation**: any civil modifications such as foundation work, bollards/gantry structures, trenching/ducting, road cutting, canopy fabrication, civil mounting structures, welding/grouting, or lane rework.
- **Electrical infrastructure upgrades**: new power lines, panel upgrades, earthing improvements, additional electrical cabling beyond last-mile connection to provided power points.
- **Network infrastructure expansion**: new network cabling across long distances, switch room expansion, new access points/towers.
- **Changes in ITUP, N4 and Sealing Interface**: Any process redesign, custom enhancements, or major configuration changes beyond agreed standard integration interfaces.

7 Assumptions & Dependencies

- **Integration readiness:** Adani and vendor IT team to confirm integration method (Rest APIs), required fields, and provide test environment + test data for validation.
- **Master data availability:** Material/vendor/transporter/plant/location codes and user roles to be shared and kept clean for mapping and reporting.
- **Process sign-off:** availability of stakeholders during the requirement gathering stage and timely sign-off on the process and workflow to be implemented.
- **Operational participation:** Stakeholders available for workshops/UAT/training; go-live support.
- **Change control:** Any changes in ITUP, N4 and sealing portal to be communicated and managed through change control.

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8 Project Plan

#	Stages	Activities	W1							W2							W3							W4							W5	
			D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23	D24	D25	D26	D27	D28	D29	D30
1	Require ment	Requirement gathering for Dashboard, Alerts & Integration																														
2		Requirement Sign-off																														
3	Design	Architecture Design																														
4		UI/UX Design																														
5		Database Design																														
6	Build	Backend Developement																														
7		Frontend Developement																														
8		API Integration																														
9	Test	Unit Testing & Remediation																														
10		SIT & Remediation																														
11		UAT & Remediation																														
12		UAT Sign-off																														
13	Relea se	DEV to PROD Cutover- Go live																														
14		Hypercare & Support																														
15	Tr ai n	Training- Train the Trainer																														

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9 Training Approach

User Manual	Online Assistance	Quarterly Training & Review Sessions	'Train the Trainer' Approach
• Provides in-depth information on solution features and functionalities. • Includes troubleshooting steps and FAQS. • Easy-to-navigate sections for quick reference.	• Tailored sessions addressing specific needs of key stakeholders. • Real-time resolution of queries and Issues. • Enhanced understanding through direct interaction. • Interactive sessions with live Q&A.	• Organized every quarter to keep users informed. • Detailed walkthroughs of new features and updates. • Ensures continuous improvement and knowledge enhancement. • Ensuring team is always at the forefront of industry standards	• Designated trainers within organization receive specialized training. • Efficient knowledge dissemination to all team members. • Ensuring users are up-to-date (Productive & proficient) • Regular feedback collection to improve the product.

Train the Trainer model

Who will be trained?

Axestrack will conduct training for select users from each user group and provide them hands on training on the application

How?

- Axestrack will conduct regular trainings as per the business requirement.
- Axestrack will provide FAQ guides and user manuals to assist in trainings

What will be they trained on?

Axestrack will provide training end to end training for the product workflow and solution approach.

Expectations from Client

- Nominate the right set of individuals who are enthusiastic and can take the learning to a boarder group
- Detailed stakeholder matrix

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10 Governance Structure

Axestrack has three layered approaches to Governance

Steering committee and value board: This layer will have the executive sponsor for the project with one-to-one mapping with the Adani's Sponsors level management. The team will meet monthly/quarterly to validate and monitor the project progress and take strategic decisions. This layer is the highest level of escalation,

Engagement & project management: This layer provides the project level monitoring and management. They will deal with the day-to-day operation issues and escalations. This layer will be responsible for the project success and excellence.

Project operations & transformation: This layer involves all the members of execution team including Technical SPOCs, functional SPOCs and SMEs. These will interact with the client project team on daily basis and work for the success of the project.

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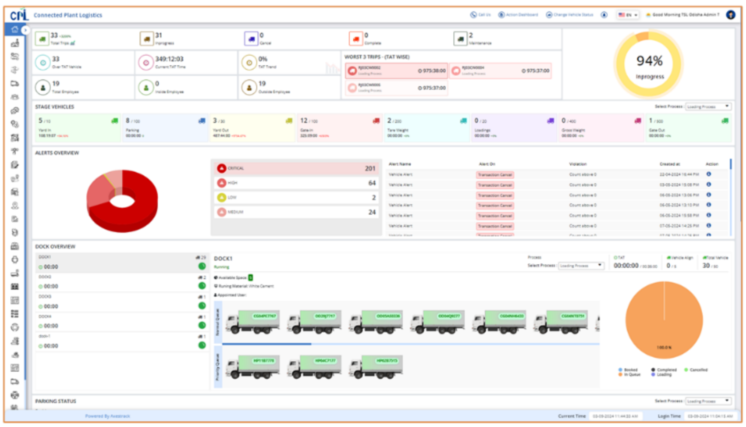
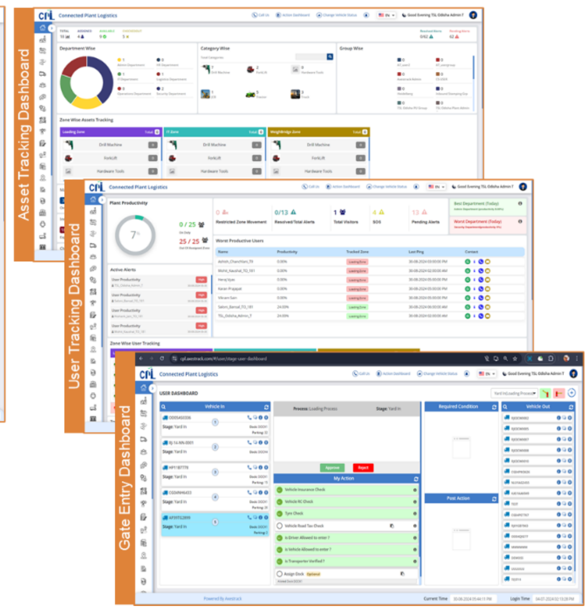
11 Overview of Connected Plant Logistics

11.1 One Stop Solution for Plant logistics & Assets Visibility

One Stop Solution for Plant & Assets Visibility…

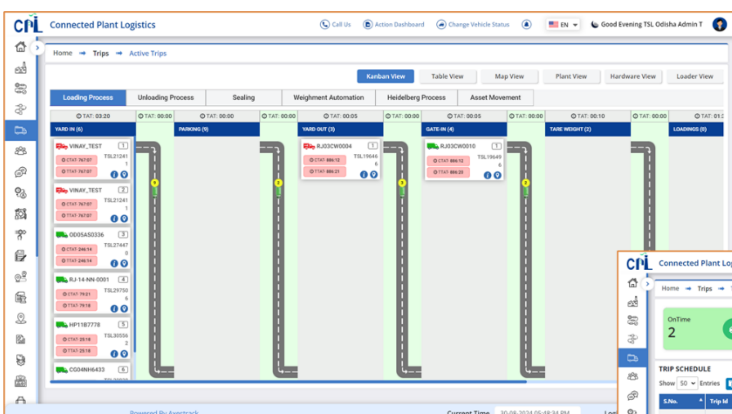




Summary Dashboard

Our dashboards **provides complete control & visibility** across all processes including Vehicle, Employee & Asset tracking

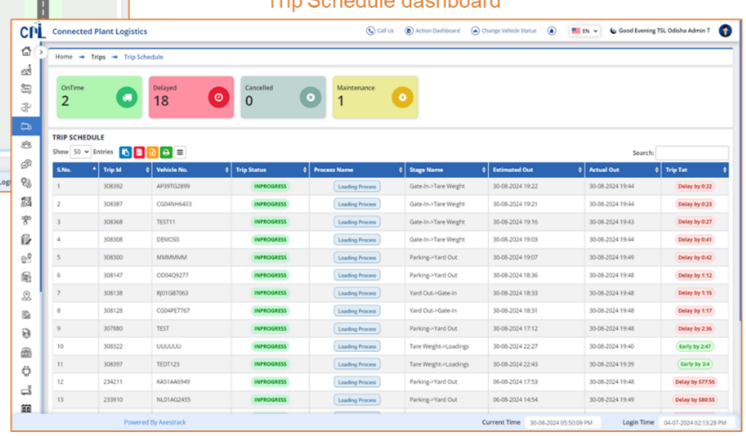
11.2 In-plant Trip Management



Trip Management dashboard

Stage-based tracking to help you measure, analyze and reduce Turnaround Time

Trip Schedule dashboard



Real Time Trip Status for actively validate bottlenecks and take optimization actions to reduce detentions and streamline the processes.

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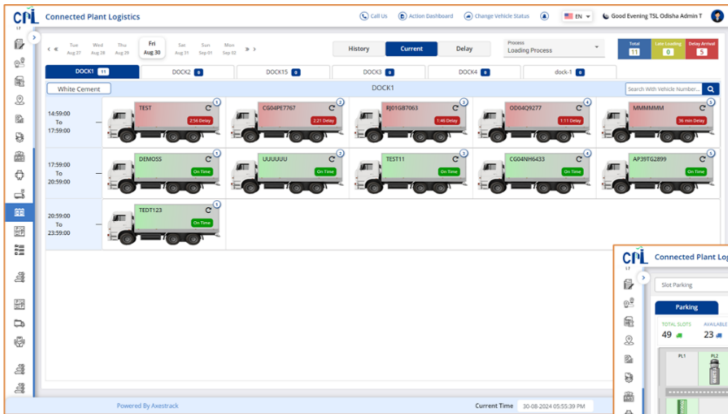
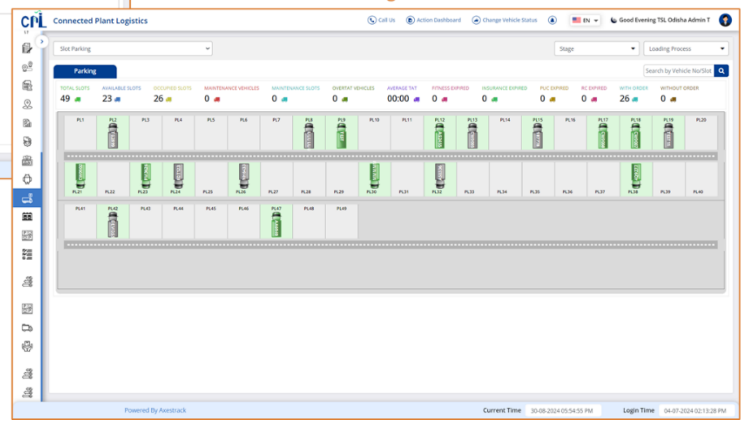
11.3 SMART Dock & Parking Management

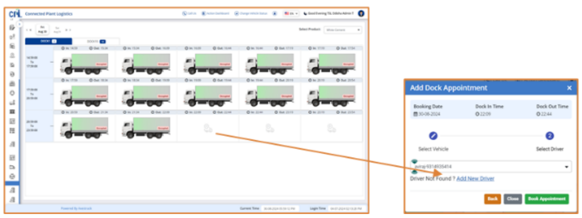


Managing parking and dockyard slots to **reduce idle time** and ensure **smooth vehicle movement** for loading and unloading materials.

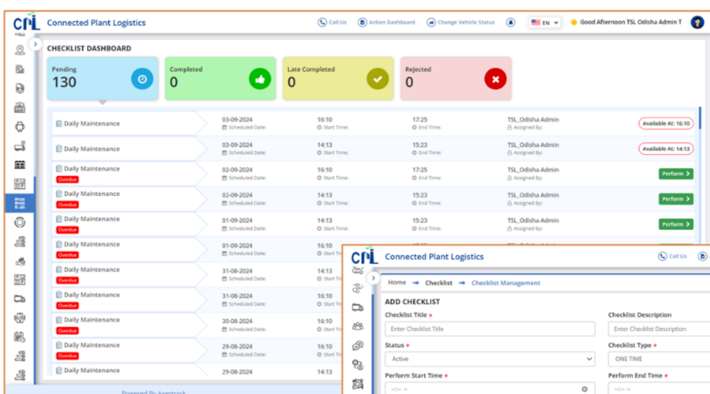
Parking dashboard



Dock Management dashboard

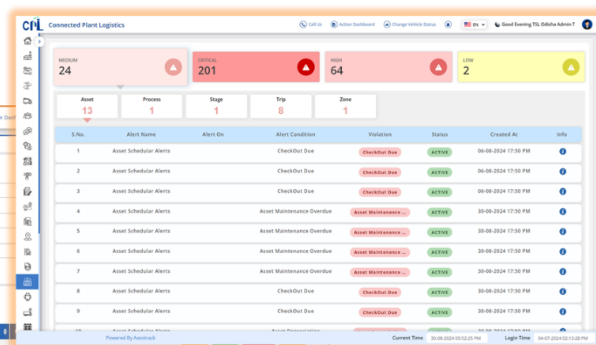


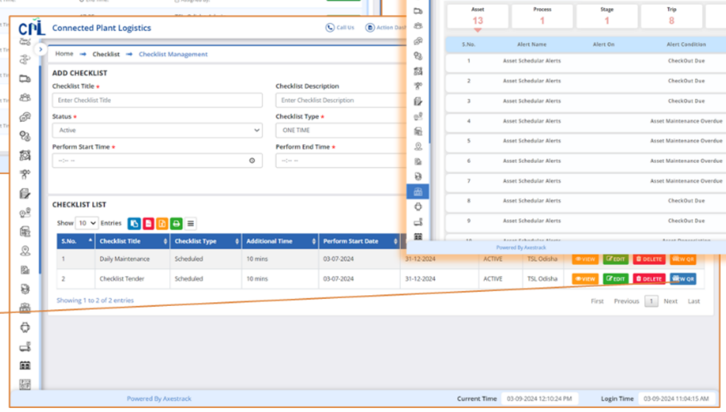
11.4 Checklist & Alerts Dashboard



Optimize Entry/Exit Time and Process for Fleets & proactive Alerts Dashboard for any violations

Alert dashboard





Checklist dashboard

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11.5 Benefits of Axestrack Connected Plant Logistics

Enhance TAT efficiency

- Automate vehicle flow across all stages to reduce delays.
- Enable real-time dock assignment and vehicle sequencing.
- Minimize congestion and ensure continuous movement.

Enable Real-Time Visibility

- Track vehicle status at every stage within the plant.
- Central dashboards for live monitoring and control.
- Reduce misrouting and improve flow predictability.

Improved Traffic management

- Regulate vehicle movement through RFID-enabled signals.
- Prevent cross-traffic conflicts and improve road safety.
- Maintain smooth lane-wise movement in operational zones.

Strengthen Compliance & Audits

- Digitally capture seal, document, and checklist validations.
- Maintain transparent, tamper-proof process logs.
- Ensure readiness for audits and internal reviews.

Manage Exceptions Proactively

- Real-time alerts for delays, deviations, or skipped stages.
- Multi-channel notifications for fast coordination.
- Improve emergency response and issue resolution.

Drive Data-Backed Decisions

- Access reports on TAT, queues, usage trends, and alerts.
- Analyze performance patterns and identify bottlenecks.
- Support planning for capacity, scaling, and resources.

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Right Information,Great Decisions